**5.** If the ratio of the orbital distance of two planets $\frac{d_1}{d_2}$ =2, what is the ratio of gravitational field experienced by these two planets?

Ans: $E_2 = 4\,E_1$

**6.** The Moon Io orbits Jupiter once in 1.769 days. The orbital radius of the Moon Io is 421700 km. Calculate the mass of Jupiter?

Ans: $1.898 \times 10^{27}$ kg

**7.** If the angular momentum of a planet is given by $\vec{L} = 5t^2\hat{i} - 6t\hat{j} + 3\hat{k}$ What is the torque experienced by the planet? Will the torque be in the same direction as that of the angular momentum?

Ans: $\vec{\tau} = 10t\hat{i} - 6\hat{j}$

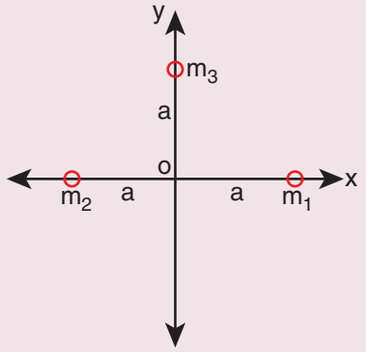
**8.** Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. Calculate the speed of each particle

Ans: $\frac{1}{2}\sqrt{\frac{GM}{R}\left(1+2\sqrt{2}\right)}$

**9.** Suppose unknowingly you wrote the universal gravitational constant value as $G = 6.67 \times 10^{11}$ instead of the correct value $G = 6.67 \times 10^{-11}$, what is the acceleration due to gravity g' for this incorrect G? According to this new acceleration due to gravity, what will be your weight W'?

Ans: g' =$10^{22}$ g, W' = $10^{22}$W

**10.** Calculate the gravitational field at point O due to three masses $m_1$,$m_2$ and $m_3$ whose positions are given by the following figure. If the masses $m_1$ and $m_2$ are equal what is the change in gravitational field at the point O?




Ans: $\vec{E} = \frac{G}{a^2}\left[(m_1 - m_2)i + m_3 j\right]$

if $m_1 = m_2$, $\vec{E} = \frac{G}{a^2}\left[m_3\,\hat{j}\right]$

**11.** What is the gravitational potential energy of the Earth and Sun? The Earth to Sun distance is around 150 million km. The mass of the Earth is $5.9 \times 10^{24}$ kg and mass of the Sun is $1.9 \times 10^{30}$ kg.

Ans: U= $-49.84 \times 10^{32}$ Joule

**12.** Earth revolves around the Sun at 30 km $s^{-1}$. Calculate the kinetic energy of the Earth. In the previous example you calculated the potential energy of the Earth. What is the total energy of the Earth in that case? Is the total energy positive? Give reasons.

Ans: K.E = $26.5 \times 10^{32}$ J

E = $-23.29 \times 10^{32}$ J

(–) ve implies that Earth is bounded with Sun

**13.** An object is thrown from Earth in such a way that it reaches a point at infinity with non-zero kinetic energy



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